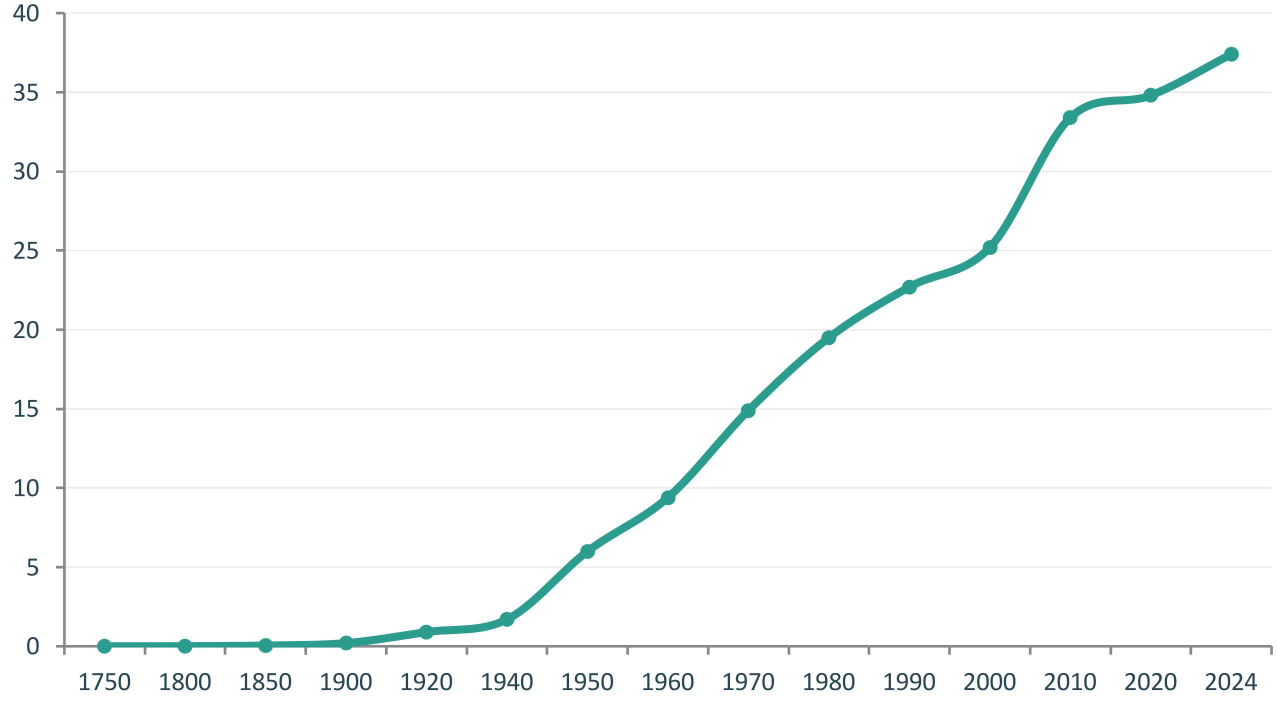
Global CO_2 Emissions: Who Emits, How Much, and What's Changing

Fossil Fuels & Industry, 1750–2024

Data: Global Carbon Budget 2025 via Our World in Data

For policy analysts, climate negotiators, and sustainability researchers

Global CO_2 Emissions Grew from 6 Bt in 1950 to Over 35 Bt Today



Pre-1950

Slow growth, negligible emissions before industrialization

1950: ~6 Bt

Post-war acceleration begins

1990: ~20 Bt

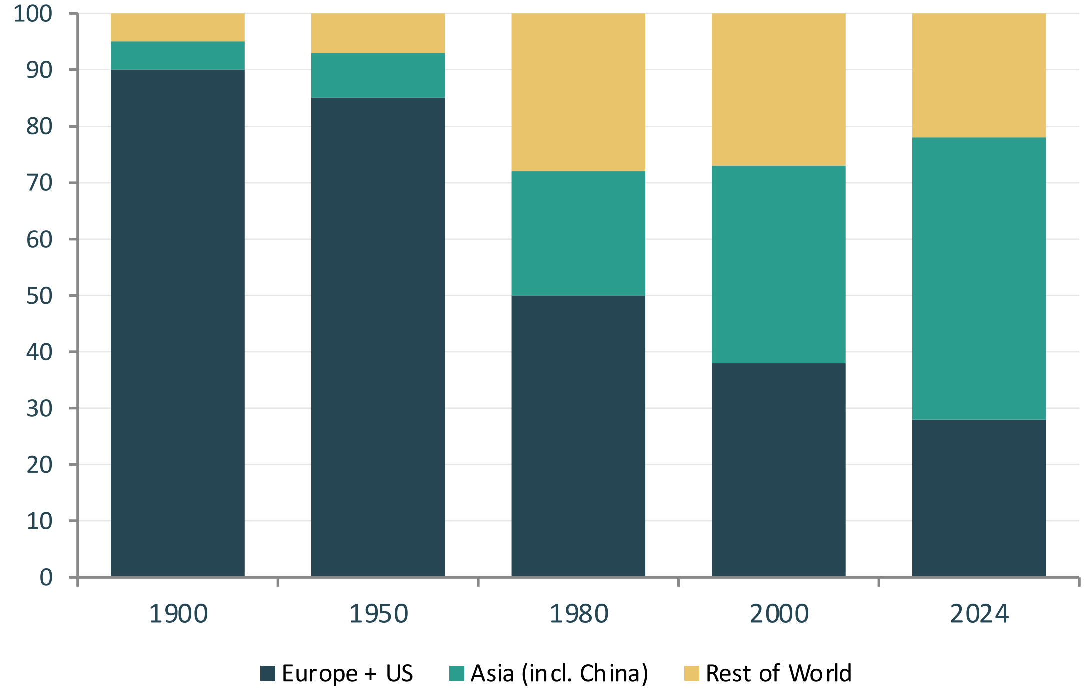
Nearly 4x increase in 40 years

2024: >35 Bt

Growth slowing but no peak yet

Source: Global Carbon Budget 2025 via Our World in Data

Europe and the US Fell from 90% of Emissions in 1900 to Under One-Third Today



1900

Europe + US: >90% of global emissions
Asia: just 5%

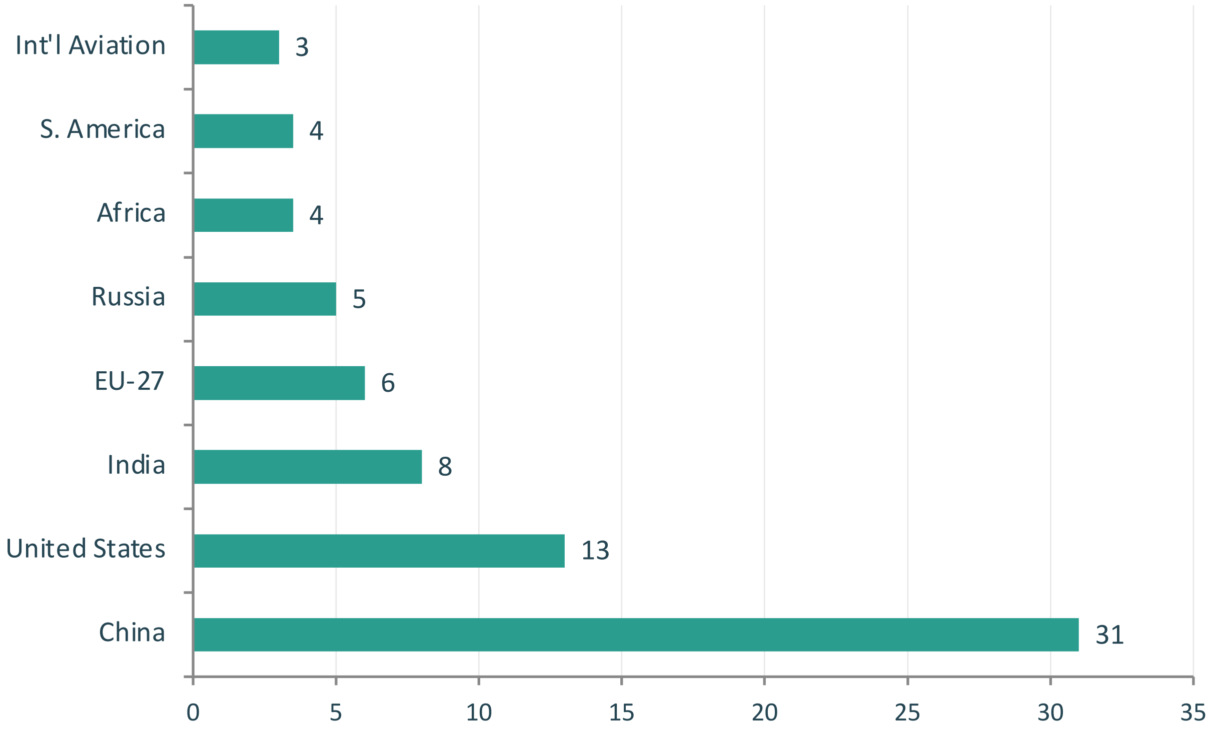
2024

Europe + US: <33% of global emissions
Asia: now ~50%

The balance of emissions shifted from West to East in just one century

Source: Global Carbon Budget 2025 via Our World in Data

China Now Emits More Than One-Quarter of Global CO_2



~31%

China's share of global CO_2 emissions (2024)

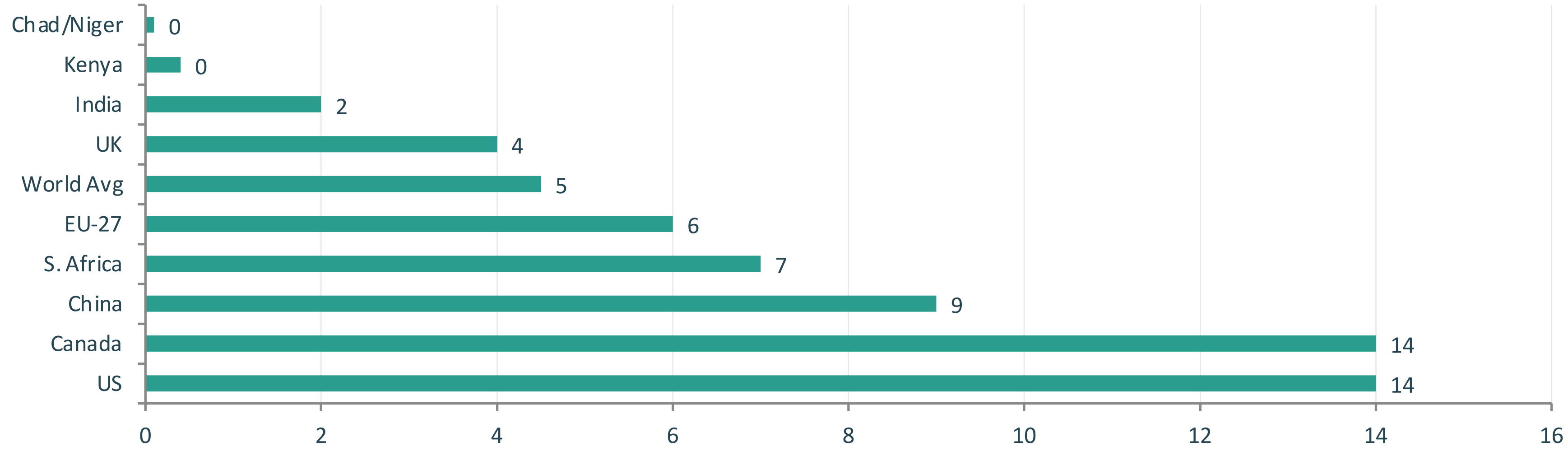
Top 4 Emitters

China: ~31%
US: ~13%
India: ~8%
EU-27: ~6%

Africa + S. America: ~7%
(comparable to int'l aviation & shipping)

Source: Global Carbon Budget 2025 via Our World in Data

Per Capita Emissions Range from 0.1 t in Chad to ~14 t in the US and Canada



150x

Gap: Chad/Niger (~0.1 t) vs US/Canada (~14 t)

Under 2 Days = 1 Year

An American emits in under 2 days what a person in Mali or Niger emits in an entire year

Similar Prosperity, Different Emissions

France & the UK emit far less per capita than the US despite similar living standards

Source: Global Carbon Budget 2025 via Our World in Data

Per Capita Emissions Vary Widely: ~14 t in the US vs. ~9 t in China and ~2 t in India

~14 t US per capita

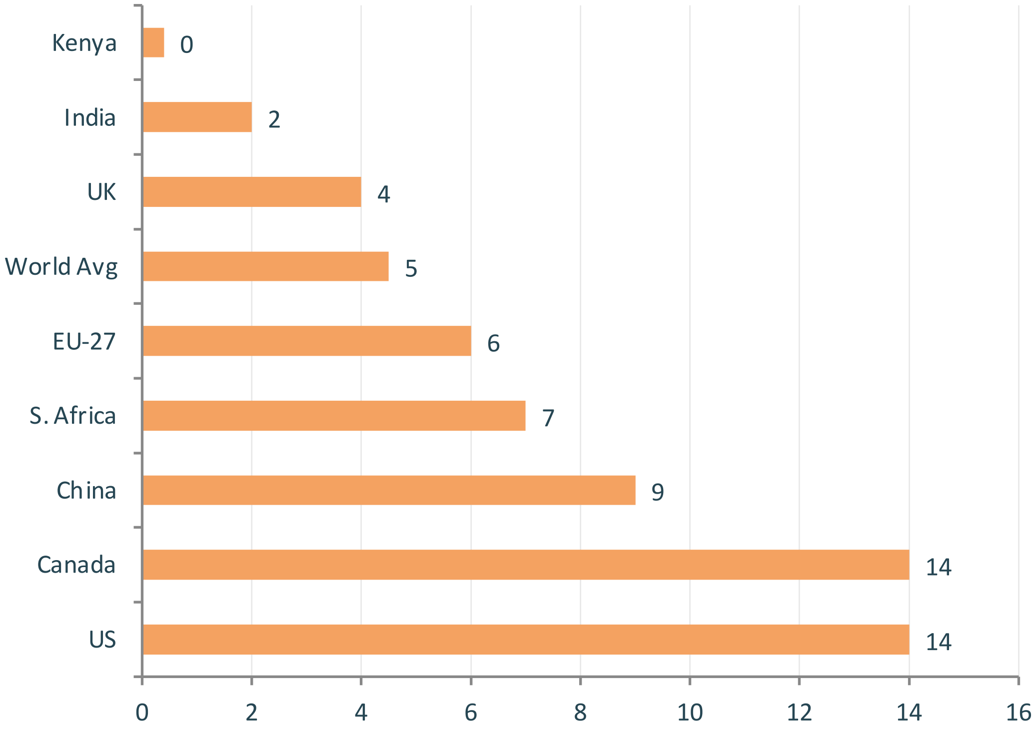
~9 t China per capita

~2 t

India's per capita emissions remain very low — well below the world average of ~4.5 t

World Average: ~4.5 t

US/Canada: ~14 t (3x world avg)
China: ~9 t (2x world avg)
EU-27: ~6 t (1.3x world avg)
UK: ~4 t (below world avg)
India: ~2 t (well below avg)

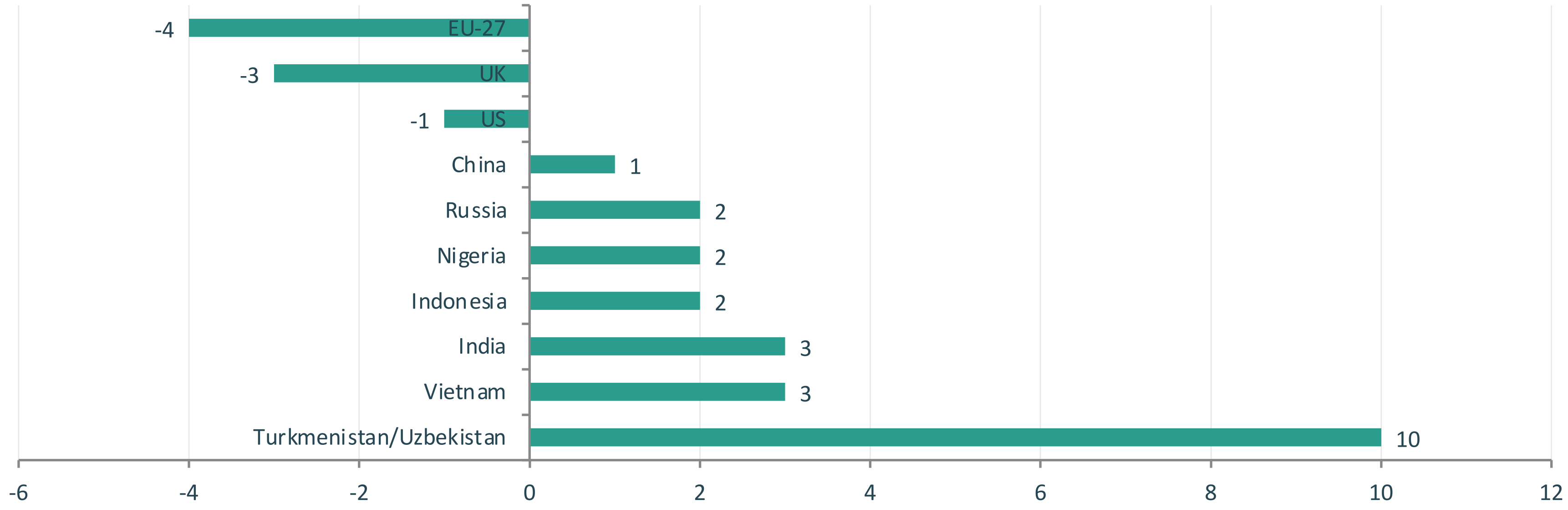


Source: Global Carbon Budget 2025 via Our World in Data

2024 Snapshot: Many Western Nations Cut Emissions While Parts of Asia and Africa Grew

Declines: US, EU-27, UK — continuing a trend of gradual reductions

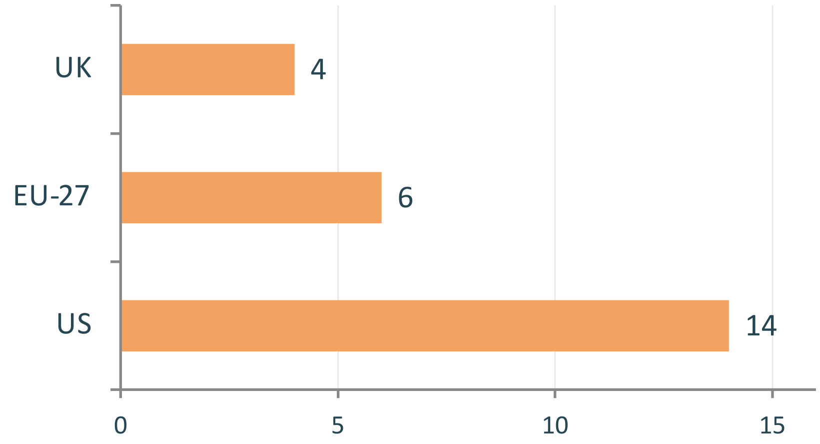
Increases: Russia, SE Asia, Africa — Central Asia exceeded +10%



Mixed Signal: Increases in Asia, the Middle East, and Africa offset some of the Western declines — global emissions still rising.

Source: Global Carbon Budget 2025 via Our World in Data

Energy Mix Explains Why France and the UK Emit Far Less Per Capita Than Germany or the US



Same Prosperity, Different Emissions

Wealthy nations with similar GDP per capita can have very different per capita emissions depending on energy choices.

Low Per Capita: Nuclear & Renewables

France (~6 t EU avg), the UK (~4 t), and Portugal have much lower per capita emissions because a higher share of their electricity comes from nuclear and renewable sources rather than fossil fuels.

High Per Capita: Fossil Fuel Dependent

Germany, the Netherlands, and the US (~14 t) rely more heavily on fossil fuels for electricity generation, driving up per capita emissions.

Energy policy and technology choices can decouple prosperity from emissions — a critical insight for climate strategy.

Source: Global Carbon Budget 2025 via Our World in Data

Key Takeaways: Responsibility Is Shared but Unequal

1. Global CO_2 emissions exceed 35 billion tonnes per year and have not yet peaked, despite slowing growth.
2. China is the largest annual emitter (~31%), but per capita emissions remain below the US (~9 t vs ~14 t).
3. Historical responsibility is dominated by Europe and the US, which accounted for >90% of emissions in 1900 but <33% today.
4. Per capita gaps of 150x persist between the richest and poorest nations — from ~0.1 t in Chad to ~14 t in the US.
5. Energy policy and technology choices can decouple prosperity from emissions, as shown by France and the UK.

Data: Global Carbon Budget 2025 via Our World in Data